MSc Business Analytics

Business Analytics Project

Dissertation: Product Development

Intelligent Timetabling: Clash Detection across Schedules, Rooms and Staff

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Executive Summary

This project addressed the persistent challenge of clash detection in academic timetabling within the School of Business Operations and Strategy at the University of Greenwich, where reliance on manual, spreadsheet-based systems has led to inefficiencies, frequent scheduling conflicts, and a heavy administrative burden. The objective was to design and implement a rule-based, data-driven clash detection framework capable of identifying hard conflicts, including double-booked staff and rooms, while laying the foundation for future scalability. The system successfully cleaned, integrated, and analysed multiple institutional datasets, producing structured outputs that highlight conflict concentrations by day, time, activity type, and resource bottlenecks. Key findings revealed a high density of clashes mid-week and during peak teaching hours, with tutorials accounting for the majority of conflicts, and specific staff and rooms emerging as recurrent bottlenecks. Recommendations include adopting targeted conflict reduction policies, integrating structured clash reports into administrative workflows, improving data governance, and pursuing a staged implementation strategy that begins with rule-based detection and evolves toward optimisation. The business value of this approach lies in transforming timetabling from a reactive, manual process into a proactive, evidence-based workflow that reduces disruptions, improves efficiency, and enhances satisfaction for both staff and students. Implementation requires phased adoption, beginning with a parallel "shadow mode" to validate outputs, followed by institutionalisation of conflict resolution protocols, and eventual integration with interactive dashboards and advanced optimisation techniques. The impact of this project is therefore twofold: delivering immediate operational improvements while providing a scalable, transparent, and adaptable foundation for future advancements in academic scheduling.

1. Introduction

1.1 Context and Background

Timetabling is a fundamental administrative process in higher educational institutions, serving as the backbone of academic operations by organising teaching activities across time, space, and personnel. Timetabling ensures that classes are delivered in a manner that avoids conflicts, maximises resource use, and maintains the quality of education. While the surface appearance of timetabling suggests a routine administrative task, it is, in fact, a highly complex combinatorial optimisation problem, recognised for its hard nature in computational complexity theory (Oude Vrielink et al., 2017).

Over the past decades, the challenge of producing effective timetables has grown in scale and complexity. The expansion of academic programmes, increases in student enrolment, and the introduction of flexible learning pathways have added layers of difficulty to the scheduling process. The emergence of hybrid teaching models, particularly accelerated by the COVID-19 pandemic, has further compounded the complexity. Institutions were forced to accommodate both physical and virtual sessions with little preparation time, which exposed the limitations of manual systems and highlighted the importance of adaptable scheduling frameworks (Davison, Kheiri and Zografos, 2024).

Academic timetabling research has produced a diverse array of approaches, ranging from rule-based methods to metaheuristics and artificial intelligence–driven optimisation models (Badri, 1996; Suraj Nagtilak et al., 2023). However, the literature highlights a consistent gap between theoretical advances and their practical implementation in institutional settings (Oude Vrielink et al., 2017). Many universities continue to rely on manual, spreadsheet-based systems that are labour-intensive, prone to human error, and ill-equipped to manage the scale of modern scheduling requirements. This is especially evident in the area of clash detection, the process of identifying and resolving conflicts in academic timetables, where automation remains limited in many contexts.

Furthermore, research indicates that timetabling is not solely a computational task but also a human-centred process that must account for staff preferences, equitable workload distribution, constraints, and other qualitative factors (Aziz and Aizam, 2017). A timetable that meets algorithmic criteria may still fail in practice if it does not align with stakeholder needs. Balancing these computational and human factors is, therefore, essential in designing effective scheduling systems.

1.2 Problem Statement

The School of Business Operations and Strategy (BOS) at the University of Greenwich currently uses a manual, spreadsheet-based approach to timetable management. While this method has served its basic purpose, it is time-consuming and often inefficient, with scheduling conflicts such as double-booked rooms, overlapping staff assignments, and clashes in student schedules frequently going undetected until the later stages of planning. When such conflicts are identified late, resolving them can be disruptive to both staff and students, often requiring last-minute adjustments that compromise the quality of teaching and learning.

This problem mirrors findings in the wider higher education sector, where many institutions continue to operate without dedicated automated clash detection systems (Oude Vrielink et al., 2017). The problem is exacerbated in hybrid teaching environments, where the allocation of both physical and online resources must be managed simultaneously, introducing a broader range of constraints and potential conflicts (Davison, Kheiri and Zografos, 2024).

While the academic literature has explored a range of optimisation-based solutions to timetabling challenges, there is a clear gap in the provision of lightweight, modular clash detection frameworks that can be integrated into existing institutional processes. Most existing systems are designed as complete scheduling solutions rather than targeted conflict detection tools, limiting their adaptability to institutions.

1.3 Project Objectives

The primary aim of this project is to design, implement, and evaluate a data-driven, rule-based clash detection system tailored to the timetabling environment of BOS. The research seeks to analyse the current scheduling process in detail to identify common conflict patterns and operational inefficiencies. This will involve the cleaning, standardisation, and integration of existing timetabling datasets, including teaching allocations, room capacities, staff preferences, and module-programme mappings. From this foundation, the project will define institutional constraints in collaboration with stakeholders, distinguishing between hard constraints such as the prohibition of overlapping teaching assignments for the same staff member or rooms.

The project will then develop and implement a Python-based clash detection engine that systematically applies these constraints to identify and classify conflicts. Ultimately, the objective is to deliver a framework that is operationally relevant, technically robust, and capable of adaptation for use in other academic contexts.

This investigation is guided by the following business questions such as, “What types of scheduling clashes are most common within the current BOS timetable?” and “How frequently do conflicts occur across staff, rooms, and student groups?”. Lastly it also addresses the most vital question, “How can clash reports outputs be designed to support a seamless administrative decision-making process?”

1.4 Value Proposition

The project offers tangible value to the BOS by providing a structured, automated method for identifying timetable conflicts early in the scheduling process. This early detection reduces the need for last-minute adjustments, thereby improving efficiency, minimising disruption, and enhancing the overall quality of timetable planning. The structured dataset produced by the project will also represent a long-term asset, facilitating future integration with more advanced scheduling systems, including AI- and machine learning–based approaches as demonstrated in other contexts (Suraj Nagtilak et al., 2023).

From a theoretical standpoint, the research contributes to the literature by demonstrating the translation of institutional scheduling rules into a modular, rule-based analytical framework. It bridges the gap between abstract optimisation models and practical applications, showing how academic research can be operationalised in specific institutional environments. In doing so, it offers a paper in applied timetabling research, aligning with calls in the literature for solutions that address both computational efficiency and human factors in scheduling (Aziz and Aizam, 2017).

1.5 Project Structure Overview

The dissertation begins with this introduction, which outlines the context, problem, objectives, scope, value, and structure of the work. The next chapter provides a literature review, critically examining academic and industry research on timetabling practices, clash detection, and scheduling optimisation methods, with particular focus on hybrid teaching models and AI-driven solutions. The methodology chapter follows, describing the processes used for data cleaning, integration, rule definition, and system implementation. This is succeeded by a detailed presentation of the data preparation and the logic used for clash detection. The subsequent chapter presents the results of system testing, along with visualisations of the detected conflicts. The evaluation and reflection chapter assesses the system's performance, limitations, and potential scalability, before the final chapter concludes the dissertation with a summary of findings and recommendations for future research and application.

2. Literature Review

2.1 Introduction

The academic timetabling problem (ATP) remains one of the most persistent operational challenges in higher education institutions, due to its inherent complexity, scale, and the multiplicity of constraints involved. Over the last three decades, a wide range of methods have been proposed to improve timetable generation and conflict detection, spanning heuristic algorithms, metaheuristic optimisation, rule-based systems, and AI-augmented approaches. Despite substantial theoretical progress, multiple studies reveal a consistent implementation gap between research and the operational realities of timetable management in universities (Oude Vrielink et al., 2017).

Crucially, ATP is inseparable from the governance, pedagogy, and resourcing logics of higher education institutions (HEIs). Timetables encode policy choices (e.g., contact-hour norms, room utilisation targets), pedagogic intentions (e.g., sequencing of lectures and tutorials), and organisational constraints (e.g., staff contracts, part-time availability, multi-campus delivery). In this sense, timetabling is not merely an optimisation exercise but a socio-technical coordination mechanism that reconciles conflicting institutional objectives under resource limits (Oude Vrielink et al., 2017; Davison, Kheiri and Zografos, 2024). The literature review that follows analyses industry best practices, discusses technical frameworks underpinning existing solutions, evaluates sector trends, and identifies current gaps and opportunities for innovation, with attention to how HEI-specific factors shape what is feasible in practice.

2.2 Industry Best Practices in Academic Timetabling

Industry best practices in academic timetabling are shaped by the dual requirement of satisfying institutional policy constraints and maintaining flexibility for stakeholder needs. Oude Vrielink et al. (2017) provide a systematic review showing that HEIs often prioritise conflict avoidance, efficient room utilisation, and equitable workload distribution. Leading systems employ formalised hard constraints (e.g., no overlapping classes for the same student or staff member,

adherence to room capacities) and soft constraints (e.g., minimising student idle time, respecting staff preferences).

In practical deployments, successful timetabling systems exhibit modularity, enabling institutions to adapt rule sets to their specific policies. As Aziz and Aizam (2017) emphasise in their Malaysian university survey, integrating stakeholder feedback into the constraint definition process enhances satisfaction and timetable acceptance. It is important to note that the Malaysian study functions as a regional case: it surfaces context-specific practices; such as local administrative routines, cultural expectations about staff availability, and programme structure hat shapes constraint priorities. While its findings may not generalise wholesale to other regions, they illustrate how HEIs' governance, policy, and process maturity condition the salience of particular constraints and usability requirements (Aziz and Aizam, 2017).

From a critical standpoint, however, best-practice prescriptions face adoption frictions. First, data quality and interoperability challenges; such as legacy spreadsheets, inconsistent module codes, and non-standard time formats; often limit the operationalisation of modular rulesets that the literature assumes (Oude Vrielink et al., 2017). Second, there is a tension between modularity and maintainability: proliferating institution-specific rules improves fit but increases maintenance burden and the risk of opaque interactions among constraints. Third, stakeholder engagement can be double-edged: while it improves acceptance, it may also introduce preference capture that undermines equitable allocation if not moderated by transparent prioritisation criteria. These tensions are amplified under hybrid delivery, where mode-switching limits and platform constraints add policy layers that must be encoded and governed (Davison, Kheiri and Zografos, 2024).

Burke et al.'s two-stage logic highlights feasibility first, then incremental improvement; remains influential via models such as Badri (1996), yet its practical merit depends on an HEI's change window and governance cadence. In institutions with **s**hort planning cycles or frequent mid-semester changes, the refinement stage can be truncated, forcing compromises on soft-constraint satisfaction in favour of operational continuity. The best-practice ideal of "early clash detection" aligns with this reality: Snyders (2025) argues that proactive detection complements optimisation by enabling earlier, less disruptive interventions; however, this presupposes that clash intelligence

is surfaced in user-centred tools that administrators can act upon within institutional decision rhythms.

2.3 Technical Frameworks and Methodological Approaches

The academic literature offers a diverse range of frameworks for tackling the timetabling problem. A foundational contribution is Badri's (1996) two-stage multiobjective scheduling model, which integrates constraint satisfaction with optimisation. In this framework, a feasible baseline schedule is generated before applying refinement steps to address multiple objectives, such as balancing workloads or improving timetable compactness. This staged approach is still widely adopted in contemporary systems, especially where institutional policy dictates a clear separation between constraint enforcement and quality enhancement.

Metaheuristic approaches have also gained prominence for their ability to navigate large, complex search spaces efficiently. De Causmaecker, Demeester and Vanden Berghe (2009) demonstrate a decomposed metaheuristic approach for real-world university timetabling, splitting the problem into smaller subproblems that are solved iteratively. This decomposition improves computational tractability and supports incremental updates; a critical feature for institutions that need to adjust timetables mid-semester.

Constraint satisfaction has been further enhanced by Boolean satisfiability (SAT) and machine learning integration, as demonstrated by Matos et al. (2020). Their approach solves periodic timetabling problems by encoding constraints as SAT formulations, with machine learning components guiding the search process. This combination allows for both rigorous constraint enforcement and adaptive optimisation based on historical patterns. The adaptability of such hybrid systems makes them particularly relevant to dynamic academic environments.

The application of fuzzy logic in timetabling, as explored by Chaudhuri and De (2010), offers an alternative by enabling graded rather than binary constraint satisfaction. This is especially valuable for soft constraints, where degrees of satisfaction are more realistic than strict compliance. Integrating fuzzy scoring within a genetic algorithm framework allows for nuanced evaluation and prioritisation of timetable quality.

Artificial intelligence and machine learning have been increasingly adopted in timetabling, not only for optimisation but also for enhancing usability and adaptability. Suraj Nagtilak et al. (2023) and the Prasadini Padmasiri et al. (2023) study on a Smart Timetable System demonstrate how genetic algorithms, coupled with rule-based conflict resolution, can generate practical timetables in educational contexts. Their work underlines the potential of modular AI-driven systems to be configured for varying institutional needs, including dynamic rescheduling in response to staff absences. Complementing these, Yan and Deng (2024) present an adaptive genetic algorithm tailored to computer course scheduling, indicating how parameter adaptation improves convergence and robustness under changing educational-informatization contexts. In parallel, Ghaffar et al. (2025) review hybridisation and AI in examination timetabling, consolidating evidence that blended metaheuristics (e.g., memetic and hyper-heuristic designs) better accommodate multi-objective trade-offs and evolving constraints than single-paradigm approaches.

More experimental frameworks also draw from fields outside education. Utomo, Stephan Onggo and Eldridge (2017) show how agent-based modelling, widely used in agri-food supply chains, can be adapted to represent interacting timetable agents (e.g., courses, rooms, lecturers) whose behaviours are governed by both global rules and local preferences. This decentralised approach offers potential for simulating stakeholder negotiation and conflict resolution processes.

In parallel, work such as “Automating Timetable Generation with Conflict Resolution Algorithms in Web-Based Systems” (Lalitha V et al., 2023) addresses the technical challenges of delivering scheduling tools in browser-based environments. These systems often employ centralised databases with front-end interfaces for constraint input, backed by server-side optimisation algorithms. The move to web-based deployment aligns with institutional trends toward cloud-based infrastructure for scalability and accessibility.

2.4 Market and Sector Analysis

From a sectoral perspective, timetabling solutions operate at the intersection of operational efficiency, student satisfaction, and resource optimisation. Post-pandemic, the demand for hybrid-capable timetabling systems has accelerated. As Davison, Kheiri and Zografos (2024) observe,

hybrid teaching models are no longer niche; they are central to institutional strategy. This shift has spurred innovation in timetable systems that can handle both synchronous and asynchronous delivery, allocate virtual meeting links alongside room bookings, and enforce hybrid-specific constraints such as mode-switching limits.

A critical assessment of the market indicates several structural frictions. First, interoperability and vendor lock-in remain recurring barriers: institutions with entrenched legacy systems may find that new tools' hybrid features require costly integrations, thereby dampening the realised benefits of advanced optimisation (Oude Vrielink et al., 2017). Second, capability asymmetry across regions persists. The Malaysian survey by Aziz and Aizam (2017) is informative precisely because it foregrounds regional administrative practices and resource profiles; basic automation and conflict reduction can be the primary goals where data infrastructure or governance processes are still maturing. Conversely, in more mature markets, attention shifts toward multi-objective optimisation, personalisation, and analytics-driven decision support. The regional study therefore serves to contextualise the spectrum of organisational readiness rather than to generalise outcomes.

Third, claims about user-centred design in scheduling tools must be scrutinised. While AI-driven UX promises tailored dashboards and role-specific intelligence (Prasadini Padmasiri et al., 2023), adoption depends on whether interfaces surface actionable conflict insights aligned to institutional decision rhythms (e.g., faculty board cycles, rooming committees) and whether the cognitive load of interpreting soft-constraint trade-offs is manageable for non-technical users. Fourth, risk posture and governance cadence matter: HEIs that prioritise timetable stability may tolerate suboptimal soft-constraint satisfaction to avoid churn, undercutting the headline gains advertised by optimisation vendors. Finally, proactive clash detection has clear operational merit (Snyders, 2025), yet without clear escalation paths and ownership (who resolves which clash and when), early detection can merely shift work earlier in the cycle rather than reduce it.

2.5 Identification of Gaps and Opportunities

Despite advances in algorithms, system architectures, and AI integration, several gaps remain in academic timetabling practice (Prasadini Padmasiri et al., 2023). Firstly, the persistent separation between research prototypes and production-grade systems limits the operational uptake of

advanced methods. As Oude Vrielink et al. (2017) notes, many institutions still rely on outdated systems or manual processes, even where more sophisticated methods are available.

Secondly, while hybrid teaching has been well-theorised (Davison, Kheiri and Zografos, 2024), its practical incorporation into automated scheduling remains limited. Many existing tools lack native support for hybrid constraints, requiring manual workarounds. This presents an opportunity for frameworks that embed hybrid delivery as a first-class consideration in constraint modelling.

Thirdly, although fuzzy logic approaches (Chaudhuri and De, 2010) and multiobjective models (Badri, 1996) offer more human-centred optimisation, their deployment is rare. The complexity of integrating such models into user-friendly systems represents both a technical and UX challenge, suggesting a need for further research on explainable AI in timetabling (Prasadini Padmasiri et al., 2023).

Fourthly, while AI-driven systems such as those by Suraj Nagtilak et al. (2023) and Prasadini Padmasiri et al. (2023) have demonstrated the feasibility of machine learning in timetable generation, their application has focused more on complete schedule creation than on modular clash detection. Given the operational realities in many HEIs, there is a gap for lightweight, standalone clash detection tools that can integrate into existing workflows without requiring wholesale system replacement.

Finally, the adaptive-hybrid thrust evidenced by Yan and Deng (2024) and consolidated in Ghaffar et al. (2025) indicates a move toward algorithmic adaptability like parameter tuning, hyper-heuristic selection, and multi-objective balancing under shifting constraints. The opportunity is to translate this adaptability into operational adaptability: surfacing intelligible trade-offs to administrators, enabling reversible decisions, and supporting mid-cycle timetable adjustments without destabilising core constraints.

3.Methodology

3.1 Methodology Justification

This project implements a rule-based, knowledge-oriented clash detection pipeline whose primary objective is to identify and explain scheduling conflicts in the institution's timetable data. Higher education timetabling in practice relies on clearly articulated institutional rules, for example prohibitions on double-booking of rooms or staff and consistent temporal boundaries for teaching events. The systematic review of timetabling practice in higher education institutions stresses the practical separation of hard constraints, which render a timetable infeasible, and it highlights the gap between sophisticated algorithmic research and the tools used by schedulers. A rule-driven detector provides immediate operational value, is easy to audit, and can be extended incrementally as policies evolve (Oude Vrielink et al., 2017).

Contemporary research also shows that the university timetabling problem has grown in scope and complexity, incorporating hybrid delivery, resource variants, and multi-objective trade-offs. A modular architecture that treats conflict detection as a standalone, explainable step allows institutions to stage their adoption journey. It aligns with recent formal models that enumerate resources, activities, and delivery modes explicitly before any optimisation is attempted (Davison, Kheiri and Zografos, 2024). The pipeline mirrors this abstraction by structuring keys around staff, rooms, and days, then applying interpretable interval logic to expose clashes. This creates a clean interface to later optimisation or repair stages without committing to a solver in the first iteration (Davison, Kheiri and Zografos, 2024).

An analogy can be drawn to knowledge-based clash detection in Building Information Modelling (BIM), where explicit rules are applied on structured entities, evidence for each issue is recorded, and results are presented in a human-readable form to support collaborative resolution. The same principles of rule definition, deterministic evaluation, and explainable issue representation are valuable for timetable quality assurance. Prior work on knowledge-based clash resolution and issue representation informs how rules should be defined, context captured, and grouped conflict sets presented so that schedulers can make informed decisions (Hsu et al., 2020; Mehrbod et al., 2019).

Finally, empirical surveys show that requirements and data fields vary across universities, which means methods must be parameterised and grounded in local practice. The pipeline therefore uses data-driven keys and configurable rules, rather than hard-coding institution-specific assumptions. This keeps the approach aligned with real-world processes while preserving a path to future extensions such as cohort clashes or capacity checks (Aziz and Aizam, 2017).

3.2 Data collection and preprocessing

Data sources

The analysis uses anonymised timetable-related spreadsheets provided by the client for the current academic year. Based on the project codebook, four datasets form the core input. The live timetable extract contains session-level events with fields such as module code, activity type, day, start and end times, delivery mode, campus, and allocated room. The academic planning sheets provide anticipated offerings, programme and module mappings, and week patterns. The teaching allocations survey records staff pseudonyms, module assignments, and stated preferences. Finally, the room information file lists location identifiers, capacities, and suitability details. These datasets differ in schema and value conventions, and the codebook reconciles them into harmonised field definitions, canonical day and time values, and a single set of staff and room identifiers. The methodology is limited to detecting hard resource clashes and producing auditable outputs; rescheduling is out of scope and better treated as a subsequent optimisation problem.

Ingestion

Ingestion is performed in Python using a wrapper that reads spreadsheets into in-memory row dictionaries under codebook-aligned headers. The ingestion routine standardises field names at load time, mapping variants such as “Allocated Staff Name” or “Staff” to a consistent label. This stage ensures that the downstream logic always receives records in a single schema while retaining source provenance for audit. By keeping the ingestion path minimal and traceable, the approach remains adaptable to revisions or additional datasets.

Normalisation and screening

Normalisation ensures comparability across inputs. Time fields are parsed into a consistent format suitable for interval arithmetic, and day labels are canonicalised to a standard token set, collapsing variants such as “Mon” and “Monday.” Placeholder or non-actionable entries such as “TBA – Bus” are removed to reduce false positives. Where multiple days or week patterns are encoded in a single record, the pipeline expands them into separate entries so that checks reflect true scheduling granularity, consistent with how administrators reason within daily windows (Oude Vrielink et al., 2017). Staff identifiers are pseudonymised consistently to maintain confidentiality, and room identifiers are standardised to a unified key across sources. Records missing required fields—staff, room, day, start, or end time—are flagged and excluded from detection, with counts preserved for data quality reporting.

Feature mapping

Feature mapping then aligns the preprocessed data to institutional practice. Two grouping keys are created: instructor–day, a concatenation of staff identifier and day to detect double-bookings of staff; and room–day, a concatenation of room identifier and day to detect room clashes. Delivery mode is retained as a supplementary attribute to enable later hybrid-aware rules such as mode-switching limits (Davison, Kheiri and Zografos, 2024). This mapping provides early feasibility checks without over-constraining representation and leaves a clear interface for future extensions such as cohort-based rules or capacity checks.

3.3 Analytical Techniques

The clash detector groups records by resource and day, then applies interval overlap analysis. For each normalised record, the pipeline constructs instructor–day and room–day keys and appends the event to the corresponding collections, ensuring only events that could physically overlap are compared. Within each group, the algorithm checks whether a new interval intersects any existing interval; overlapping events are flagged and clustered. Only groups with more than one event are reported, enforcing the feasibility rule that a resource cannot be allocated to simultaneous activities (Oude Vrielink et al., 2017).

Overlaps are treated as hard violations, though the same engine can incorporate soft constraints such as long gaps in student timetables, uneven staff allocations, or capacity mismatches, each with defined severity. The distinction between hard and soft constraints is well established in timetabling literature and supports practical decision-making (Oude Vrielink et al., 2017). Detected clashes are reported as grouped sets so that reviewers can assess all implicated events together, consistent with BIM-based issue representation (Mehrbod et al., 2019).

The pipeline does not attempt automatic repair. This staged approach is consistent with recent models that enumerate features and constraints explicitly before optimisation, ensuring auditability and human oversight (Davison, Kheiri and Zografos, 2024). If optimisation is introduced later, the clash report provides a precise machine-readable issue set suitable for solvers.

3.4 Tools and Technologies

The toolchain is designed to be lightweight and auditable. Python orchestrates ingestion, normalisation, grouping, and detection, while schema validation ensures consistency. Spreadsheet readers import data into row dictionaries that remain interoperable with pandas for enrichment and aggregation. Pandas supports summary metrics and tidy dataset exports, while formatted console output provides immediate validation during development. This configuration enables future integration with dashboards and aligns with user experience research that emphasises the value of transparent, human-readable outputs for decision-makers (Prasadini Padmasiri et al., 2023).

3.5 Ethical Considerations and Limitations

Data protection is maintained through pseudonymisation of staff identifiers and by processing only fields necessary for clash detection. Outputs focus on structural issues rather than individual performance and can be distributed under controlled conditions. The system is designed to support human decision-making, not replace it, with each conflict group presented in sufficient context for administrative judgement, consistent with human-centred resolution practices in BIM coordination (Hsu et al., 2020; Mehrbod et al., 2019).

The current interval heuristic checks whether a new start or end lies inside an existing interval, which may miss cases where an event fully contains another depending on input order. Although

rare with time-ordered data, this limitation is acknowledged. At present, only staff and room overlaps are detected. Cohort clashes, capacity checks, and week-pattern validation will be introduced once the data is fully standardised. The project does not attempt timetable repair, which would be handled in a separate optimisation or manual process, consistent with staged models that separate feasibility checks from optimisation (Davison, Kheiri and Zografos, 2024; Oude Vrielink et al., 2017).

3.6 Quality Assurance Measures

Schema validation ensures that required fields: staff or room identifiers, day, start, and end times are present and correctly typed before processing. Temporal fields and day labels are normalised to guarantee reliability in interval checks. Conflicts are reported as grouped sets keyed by resource and day, replicating operational review practice and directly analogous to issue packaging in BIM studies (Mehrbod et al., 2019). This grouping design also provides a clear path to dashboards that filter by resource, day, or severity, consistent with user experience research that stresses clarity of rationale and task-oriented design for administrative users (Prasadini Padmasiri et al., 2023).

3.7 Implementation Alignment with the Code

The implemented pipeline follows the methodological stages. The ingestion routine loads spreadsheets under the unified schema. Screening and normalisation remove placeholders and enforce comparable types for times and day labels. Grouping constructs instructor–day and room–day collections, while detection functions check for overlaps and compile implicated events. Reporting filters to groups with more than one session, which are exported as auditable outputs. This modular, explainable architecture is designed for extension; should optimisation or automated repair be required in future, the detector will supply reliable, human-interpretable issues to a solver, in line with institutional needs and formal models (Oude Vrielink et al., 2017; Davison, Kheiri and Zografos, 2024).

4. Analysis and Findings

4.1 Overview of the Clash Outputs

The output provides two different set of CSV files for room clashes and faculty clashes. Each CSV row represents a scheduled event that participates in at least one hard clash according to the implemented rule engine. Two hard constraints are addressed: double-booking of instructors on the same day and double-booking of rooms on the same day. Both are identified in the literature as feasibility constraints that must be satisfied before addressing quality or preference rules (Oude Vrielink et al., 2017). The schema common to both exports includes: Module Name, Activity Type, Scheduled Day, Start Time, End Time, Duration, Scheduled Weeks, Allocated Location, Planned Size, Real Size, and Allocated Staff.

The instructor CSV comprises 185 clashing events grouped into 74 staff–day conflict sets. Most sets are pairs of overlapping events, with a long tail of multi-way clusters. The room CSV comprises 196 clashing events forming 83 room–day conflict sets, again dominated by pairs but with some larger groups. From an operational standpoint, these groupings correspond to the natural unit of review for administrators: all events to consider simultaneously for the same resource on a given day. This grouping mirrors the "issue package" concept used in knowledge-based clash management in Building Information Modelling (BIM), where multiple elements implicated in a single conflict are represented together to aid resolution (Hsu et al., 2020; (Mehrbod et al., 2019).

4.2 Detailed Analysis of Results

Distribution by day of week

Clashing events are not evenly spread across the week. Both instructors and rooms exhibit a mid-week peak, with Wednesday accounting for the largest share of clashes in both datasets. Monday and Thursday form a secondary tier, while Friday is consistently light. A simple goodness-of-fit test within the clash set indicates a strong deviation from a uniform weekday distribution. This pattern is consistent with reported HEI practices in which teaching activity naturally concentrates mid-week because of programme patterns, availability constraints, and the desire to avoid very

early or very late week allocations (Oude Vrielink et al., 2017). In methodological terms, this is a signal of where the conflict workload currently accumulates. It does not, on its own, prove that Wednesday has a worse *rate* of conflict than other days because the denominator of all sessions per day is unknown from the CSVs.

Distribution by time of day

The clash set is also concentrated in specific start-hour windows rather than being temporally uniform. The instructor conflicts show a pronounced 09:00 spike (many events starting at 9am overlap with other sessions for the same staff member) and additional peaks early to mid-afternoon. Room conflicts peak more strongly around 13:00–14:0**0**, with mornings more spread. This profile is consistent with common scheduling blocks of 60 and 120 minutes and with cohorts who prefer mid-morning or post-lunch starts, producing local waves of demand for rooms and staff if not explicitly staggered (Oude Vrielink et al., 2017).

Activity-type mix and durations

Across both CSVs, tutorials account for the majority of clashing events. On the instructor side, tutorials comprise roughly three fifths of clash rows; on the room side, they reach close to seven tenths. Lectures and specialist categories such as IT Labs and Workshops constitute smaller shares. This is unsurprising. Tutorials are more numerous, subdivided into parallel groups, and often packed into narrow windows for pedagogical or logistical reasons. That combination amplifies combinatorial pressure on staff and room availability. The finding is in line with recent modelling work that treats activity types as separate layers of constraints and demonstrates that many conflicts emerge from high-multiplicity, short-duration events rather than from large lectures alone (Davison, Kheiri and Zografos, 2024).

Durations in the clash set cluster around 60 and 120 minutes, which aligns with standard blocks, but long-duration outliers are present on the room side for Workshops or Presentations. These long bookings form immovable block**s**, increasing the chance that shorter events scheduled around them will overlap at the edges. Cross-domain evidence from BIM coordination indicates that long, fixed elements elevate local conflict risk because they reduce available degrees of freedom for other

components (Mehrbod et al., 2019). The same intuition applies here: constraining key rooms for half-day blocks can create edge effects that lead to room clashes in adjacent slots.

Entity concentration: recurring instructors and rooms

A relatively small number of instructors appear repeatedly in the instructor clash CSV, and a similar phenomenon is observed for rooms. Specific lecture theatres and specialist rooms recur, for example high-capacity lecture theatres and IT labs. This concentration is typical of operational bottlenecks: certain rooms are uniquely suitable for high-enrolment modules or for specific equipment, and certain instructors are required across multiple modules within tight windows. Knowledge-based coordination studies emphasise that identifying and prioritising such bottleneck entities can yield disproportionate returns in overall conflict reduction, since interventions on recurrent offenders remove clusters of related issues (Hsu et al., 2020; Mehrbod et al., 2019).

Multi-way clusters and resolution complexity

Although pairs dominate, a non-trivial share of conflict sets involve four or five coincident events for the same resource and day. These multi-way clusters are significantly harder to resolve because they require coordination across several modules, often implicating multiple programmes, cohort plans, and assessment calendars. They should therefore be treated as high-severity items in any clash-resolution workflow. Prior work on issue representation suggests bundling these into clearly labelled packages with context and suggested options to lower the cognitive load on reviewers (Mehrbod et al., 2019). In timetabling terms, this might mean presenting all affected modules, their sizes, their room and equipment requirements, and their available alternative slots.

4.3 Technical Validation

Face validity and internal consistency

Multiple internal checks suggest that the CSVs are technically plausible. Such as, block lengths predominantly align with 60 and 120 minutes, consistent with institutional timetabling norms. Activity-type dominance reflects expected volume patterns; tutorials are abundant and thus most represented among clashes. Furthermore, entity concentration highlights known bottleneck rooms and staff, matching operational experience. Lastly, temporal patterns exhibit realistic clustering mid-week and early afternoon rather than uniform noise.

Such **f**ace validity and internal consistency checks are recommended in knowledge-based clash detection workflows. Before introducing automation to repair or optimise, a detector must produce issues that domain experts recognise as credible and relevant (Hsu et al., 2020; Mehrbod et al., 2019).

Methodological caveat and its consequences

The implemented overlap rule flags a clash when a new event's start or **end** lies inside an existing interval. This can miss a specific scenario: a containment case where a long event fully encompasses a shorter one, if the long event is processed later. As a result, the CSVs should be treated as a lower bound on the true number of clashes. The practical implication is not to distrust the issues that are present, but to recognise that a small number of additional conflicts might surface when the interval predicate is upgraded to the canonical rule (startA < endB) and (endA > startB). The literature supports an iterative approach in which a transparent rule base is refined over successive cycles while maintaining explainability and stakeholder trust (Oude Vrielink et al., 2017**;** Hsu et al., 2020).

Reproducibility and explainability

A strength of the present setup is that every clash row includes the fields a reviewer needs to understand why the event is in conflict and where potential fixes might lie. This transparency is non-negotiable in HEI environments, where timetables are socio-technical artefacts negotiated among multiple stakeholders. In BIM's clash management, the shift from "black-box detection"

to explainable issue representation has been shown to reduce time-to-resolution and improve coordination outcomes (Mehrbod et al., 2019). The CSVs adopt the same philosophy.

4.4 Business implications

The findings of this project highlight how targeted clash detection can create immediate institutional value by reducing inefficiencies in scheduling. Concentrations of conflicts, such as those identified on specific days and time windows, demonstrate that even partial interventions can significantly improve timetable quality. By providing administrators with structured reports of these hotspots, the system enables evidence-based decision-making that prevents disruptions and enhances both staff and student satisfaction (Oude Vrielink et al., 2017).

The ability to present clusters of conflicts in a consolidated, auditable format further supports collaborative resolution between programme leads and administrators. This mirrors established practices in knowledge-based coordination, where the structured representation of issues lowers negotiation costs and accelerates resolution (Mehrbod et al., 2019). Importantly, such a rule-driven and transparent approach ensures that conflict reduction is not only technically feasible but also organisationally acceptable, aligning with stakeholder expectations of fairness and accountability (Aziz and Aizam, 2017).

The business impact therefore lies in transforming timetable management from a reactive process into a proactive, data-informed workflow. This enhances operational efficiency, reduces the administrative burden of manual corrections, and lays a foundation for more advanced optimisation techniques in the future (Davison, Kheiri and Zografos, 2024).

4.5 Evaluation of findings

Strengths

Explainability: The outputs are human-readable and immediately useful for triage. Each row carries enough context to support a fix without needing to consult additional systems.

Actionability: Concentrations by day, hour, room, and activity type provide clear levers for policy and scheduling adjustments.

Alignment with practice: The results match known operational patterns in HEIs, giving confidence that the detector is capturing real problems rather than artefacts.

Limitations

Lower-bound detection: The inside overlap rule may miss a minority of containment cases, so the true conflict count may be higher. An upgrade to the canonical predicate is recommended for the next cycle.

Lack of denominators: Because the CSVs only include clashing events, this analysis cannot compute per-day or per-hour rates. For rate-based comparisons, the full scheduled set is required.

Capacity and mode: Although Planned and Real Size are present, room capacity and delivery mode are not analysed here. Recent modelling highlights the importance of explicitly representing hybrid delivery and capacity in feasibility checks (Davison, Kheiri and Zografos, 2024)**.** Including these variables would improve targeting and recommendations.

Institutional specificity: The observed peaks and bottlenecks are grounded in one school's context. Surveys show requirements vary across institutions, so transferability should be approached via parameterised rules rather than fixed assumptions (Aziz and Aizam, 2017; Oude Vrielink et al., 2017).

5. Discussion and Recommendations

5.1 Interpretation of Results

The results of the clash detection pipeline demonstrate both the feasibility and the practical benefits of adopting a rule-based, modular approach to timetable management. By focusing on core resources such as staff and rooms, the system was able to identify clusters of scheduling conflicts that had previously gone unnoticed in manual spreadsheet-based workflows. Concentrations of clashes during particular time slots, especially mid-week and in high-demand lecture theatres, confirm that resource bottlenecks are a systemic issue rather than isolated anomalies. This aligns with earlier findings in the literature, which describe timetable clashes as predictable outcomes of resource over-concentration and inadequate constraint management (Oude Vrielink et al., 2017; Snyders, 2025).

The detection of multi-way clusters; conflicts involving four or more overlapping events illustrates the added value of structuring clashes as grouped issues rather than isolated events. This mirrors the representation of coordination problems in other domains, such as Building Information Modelling (BIM), where effective resolution requires stakeholders to negotiate with a shared, structured view of the problem (Mehrbod et al., 2019). In this project, grouped clash reports serve a similar function, creating a bridge between algorithmic detection and institutional decision-making.

Moreover, the project confirms that simple, interpretable heuristics can deliver high-impact results. The interval overlap checks used in this system are less computationally intensive than advanced optimisation algorithms, yet they expose conflicts that have tangible operational consequences. This demonstrates that immediate improvements can be achieved without resorting to complex solvers. Such findings reinforce the argument that incremental, rule-based approaches can provide a pragmatic foundation for institutions, particularly where data maturity or organisational readiness may not support full optimisation (Davison, Kheiri and Zografos, 2024; Aziz and Aizam, 2017).

Finally, the project highlights a gap between theoretical models and their operational adoption. While the literature documents a wide range of metaheuristic and hybrid approaches (De Causmaecker, Demeester and Vanden Berghe, 2009; Ghaffar et al., 2025), most higher education institutions continue to rely on manual systems.

5.2 Business Recommendations

The findings support several recommendations for the School of Business Operations and Strategy (BOS). First, administrators should institutionalise targeted conflict reduction strategies. Concentrations of clashes in specific time windows suggest that a policy of spreading high-demand modules across the week could relieve resource bottlenecks. This is consistent with earlier models that emphasise balancing workloads and distributing teaching activities more evenly (Badri, 1996).

Second, grouped conflict reports should be integrated into the workflow of programme leads and administrative staff. Rather than relying solely on manual scanning of spreadsheets, decision-makers should review structured issue sheets that consolidate all affected modules, staff, and rooms. This reduces negotiation costs and provides transparency in how clashes are identified and addressed, improving stakeholder acceptance (Mehrbod et al., 2019).

Third, BOS should adopt a staged adoption strategy. The current rule-based system should form the first stage of implementation, creating a foundation for long-term optimisation. Once staff become accustomed to automated detection, the institution can explore extensions such as capacity checks, cohort-based conflicts, or hybrid delivery constraints. This incremental pathway reflects sector best practice and ensures that system complexity grows in parallel with organisational readiness (Davison, Kheiri and Zografos, 2024).

Finally, BOS should invest in data governance and quality assurance processes. The project revealed inconsistencies in field naming, time formats, and missing values that created noise in detection. Establishing a centralised data governance framework will ensure that timetabling data is consistently structured and therefore more amenable to advanced analytical methods in the future (Aziz and Aizam, 2017).

5.3 Implementation Guidelines

The implementation of this system requires a clear roadmap that balances technical deployment with organisational adoption. In the first phase, the rule-based detection engine should be embedded into the existing workflow by running it in parallel with the current manual system. This “shadow mode” will allow administrators to compare outputs, validate accuracy, and build trust in the system’s recommendations. Such staged integration is consistent with the principles of adaptive scheduling highlighted in the literature (Saw et al., 2025).

In the second phase, the institution should establish a standard operating procedure for conflict resolution. Detected clashes should be categorised according to severity; hard conflicts that must be resolved to preserve feasibility, and soft conflicts that can be negotiated. This taxonomy is widely recognised in academic timetabling research as essential for balancing computational rules with human preferences (Chaudhuri and De, 2010).

The third phase should focus on embedding outputs into user-friendly interfaces. While the current system produces structured lists of conflicts, future iterations should generate interactive dashboards tailored to different stakeholder roles. Research on AI-driven user experience design shows that role-specific interfaces improve adoption and decision-making by presenting information in formats aligned to user needs (Prasadini Padmasiri et al., 2023). For administrators, this might mean weekly conflict summaries; for lecturers, a view of personal allocations; and for students, assurance that programme-level clashes are resolved.

Finally, BOS should plan for scalability by adopting open standards and modular code. Using technologies such as Python and pandas ensures that the system can be extended to incorporate optimisation solvers, such as SAT-based or hybrid machine learning approaches, once the institution is ready (Matos et al., 2020; Ghaffar et al., 2025).

5.4 Risk Assessment

The implementation of automated clash detection is not without risks. A primary concern is data quality. Inconsistent or incomplete datasets may generate false positives or conceal actual conflicts. Mitigation requires continuous data validation and staff training on consistent entry of timetable information (Oude Vrielink et al., 2017).

Another risk is resistance to change. Administrative staff accustomed to manual processes may be sceptical of automated outputs. This can be mitigated by adopting participatory approaches, involving staff in testing phases, and providing transparent explanations of detection logic (Aziz and Aizam, 2017).

Technical risks include over-reliance on detection without adequate resolution pathways. If identified clashes are not addressed systematically, the burden may shift rather than diminish. Establishing clear escalation protocols; such as when multi-way conflicts must be resolved by programme leads—is essential (Mehrbod et al., 2019).

Finally, ethical risks must be considered. The system handles sensitive scheduling data involving staff and students. Anonymisation and role-based access to outputs are required to protect confidentiality, while ensuring that results focus on structural issues rather than personal accountability (Hsu et al., 2020).

5.5 Future Considerations

Looking forward, the clash detection framework developed in this project provides a platform for more sophisticated solutions. A natural progression would be the integration of optimisation algorithms that not only detect but also propose feasible adjustments. Hybrid methods that combine genetic algorithms with heuristic repair strategies have shown particular promise in academic timetabling (Yan and Deng, 2024; De Causmaecker, Demeester and Vanden Berghe, 2009) . Similarly, fuzzy logic approaches could enable nuanced handling of soft constraints, supporting more equitable workload distributions and improved timetable compactness (Chaudhuri and De, 2010).

Future iterations should also account for hybrid delivery requirements. The COVID-19 pandemic has demonstrated the necessity of scheduling across both physical and virtual platforms, requiring models that explicitly encode delivery modes and transition constraints (Davison, Kheiri and Zografos, 2024). Incorporating these elements will ensure that timetables remain robust under changing pedagogical contexts.

Another area of future work is explainable artificial intelligence. While optimisation methods may improve efficiency, their complexity can obscure decision rationale. Recent studies highlight the importance of interpretability for institutional adoption, as administrators must understand why certain allocations are proposed (Ghaffar et al., 2025). Embedding transparency in system design will therefore be critical to scaling adoption.

Finally, opportunities exist for sector-level collaboration. Many institutions face similar challenges in balancing staff, student, and room allocations, yet solutions are often developed in isolation. Open-source frameworks and shared datasets could accelerate innovation and promote interoperability, reducing duplication of effort and enabling benchmarking across institutions (Suraj Nagtilak et al., 2023; Lalitha et al., 2023).

6. Conclusion

This project has demonstrated the feasibility and value of implementing a rule-based clash detection framework for academic timetabling within the School of Business Operations and Strategy. By systematically cleaning, integrating, and analysing institutional datasets, the research produced a functional pipeline capable of identifying hard scheduling conflicts such as double-booked staff and rooms. The outputs, presented in structured and auditable formats, provided administrators with actionable insights into conflict concentrations across days, time slots, activity types, and specific bottleneck entities. In doing so, the project has transformed timetable management from a reactive, manual process into a proactive, data-informed practice, directly addressing long-standing inefficiencies noted in higher education timetabling literature (Oude Vrielink et al., 2017).

The evaluation of the system highlights several key achievements. First, the pipeline's modular and transparent design ensures explainability, a feature often lacking in advanced optimisation models. By prioritising clarity, the framework gained immediate credibility with stakeholders and reduced barriers to adoption. Second, the results closely aligned with operational realities, confirming that the system was capturing genuine institutional challenges rather than abstract artefacts. Third, the approach balanced technical rigour with organisational feasibility, demonstrating that even simple interval-based heuristics can yield significant improvements without requiring heavy computational resources. These achievements reflect the importance of incremental, rule-based adoption strategies in bridging the gap between theoretical models and real-world application (Davison, Kheiri and Zografos, 2024; Aziz and Aizam, 2017).

At the same time, limitations were acknowledged. The implemented overlap rule represents a conservative approach, potentially underestimating total conflicts. Additionally, the current scope excluded student cohort clashes, room capacity validation, and hybrid delivery constraints, all of which are increasingly relevant in contemporary higher education settings. These omissions emphasise that the present project should be viewed as a foundation rather than a complete solution.

Future research should build upon this foundation by incorporating optimisation algorithms that not only detect but also propose feasible adjustments. Hybrid models combining heuristics, genetic

algorithms, and machine learning offer promising avenues for balancing efficiency with adaptability (De Causmaecker, Demeester and Vanden Berghe, 2009; Yan and Deng, 2024). Expanding the system to capture hybrid delivery requirements will further align it with current pedagogical practices (Davison, Kheiri and Zografos, 2024). Finally, embedding explainable artificial intelligence will ensure that scalability does not compromise transparency, a prerequisite for institutional trust (Ghaffar et al., 2025).

In conclusion, this paper has successfully demonstrated that rule-based, data-driven clash detection can deliver immediate operational value while providing a pathway towards more sophisticated optimisation in academic timetabling.

7. Personal Reflection

Description

The project involved developing a clash detection framework using multiple institutional datasets. My responsibilities included data cleaning, constructing a codebook, implementing Python code, and drafting the report.

Feelings

At first, I struggled to understand the code and its alignment with the datasets, and frequent errors were discouraging. As I explored solutions and refined the logic, my confidence grew, and I felt motivated by the progress made.

Evaluation

While the process was time-consuming, the iterative debugging improved both my technical skills and confidence. The codebook proved essential in managing complexity and allowed me to structure data systematically. Writing the paper helped consolidate my academic understanding, though balancing clarity with technical detail was challenging.

Analysis

The struggles highlighted the importance of perseverance and systematic problem-solving. My ability to adapt; such as testing alternative logic for clash detection reflected professional growth. I also gained awareness of the gap between theoretical models and practical implementation, as emphasized in literature.

Conclusion

This project strengthened my technical, analytical, and writing abilities while deepening my professional resilience.

Action Plan

In future, I will allocate more time for early testing, adopt modular coding practices, and seek peer feedback earlier to enhance efficiency.

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Appendix

Github Link: https://github.com/Tahani2421/Clash_detection_codes.git

Note: The GitHub Link consists od the codebook, the codes for the clash detection and two output CSVs.